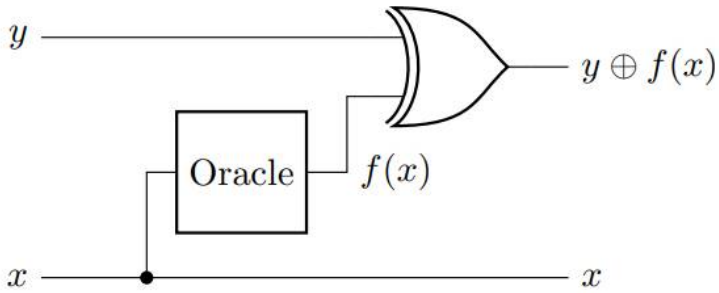
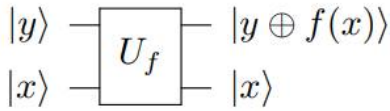
Quantum Oracle

7.1.3 Quantum Oracles

An oracle is simply a *boolean function*, meaning a function that acts on bits. Then, it can be defined using a truth table, and it can be constructed using logic gates. For it to be a quantum oracle, however, it needs to be reversible. Fortunately, from Section 1.5.4, we can turn it into a reversible circuit by XORing its output with an extra bit. For example, if the function is $f(x)$, the reversible circuit is



Since this entire circuit is reversible, it is a quantum gate. Let us call the gate U_f to emphasize that it is unitary. We can draw it as



That is, the quantum oracle U_f acts as

$$|x\rangle|y\rangle \xrightarrow{U_f} |x\rangle|y \oplus f(x)\rangle.$$

Note we can find $f(0)$ and $f(1)$ by setting $y = 0$. That is,

$$|0\rangle|0\rangle \xrightarrow{U_f} |0\rangle|0 \oplus f(0)\rangle = |0\rangle|f(0)\rangle,$$
$$|1\rangle|0\rangle \xrightarrow{U_f} |1\rangle|0 \oplus f(1)\rangle = |1\rangle|f(1)\rangle.$$

The extra qubit $|y\rangle$ is called an *answer qubit* or *target qubit*, and $|x\rangle$ is called the *input qubit*.

Think of a **quantum oracle** as a *reversible black-box function*. The twist is: quantum circuits must be reversible, so we can't just do $f(x)$ directly.

🔁 How we make it reversible

We introduce an extra qubit y and define the oracle as:

$$U_f : \ |x\rangle|y\rangle \ \rightarrow \ |x\rangle\,|y \oplus f(x)\rangle$$

👉 Instead of *overwriting*, we **XOR the output into** y.

⚙️ What's really happening

- Input qubit: $|x\rangle$
- Target (answer) qubit: $|y\rangle$
- Output:
  - x stays unchanged
  - y flips **only if** $f(x) = 1$

🎯 Special trick (very important)

Set $y = 0$:

- $|0\rangle|0\rangle \rightarrow |0\rangle|f(0)\rangle$
- $|1\rangle|0\rangle \rightarrow |1\rangle|f(1)\rangle$

👉 So the second qubit directly becomes **the function output**.

🧠 Why this matters

- Keeps the operation **reversible (unitary)** ✅
- Works inside quantum circuits
- Lets algorithms (like **Deutsch / Shor**) query $f(x)$ in superposition

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🏷️ Terminology

- $|x\rangle$: **input qubit**
- $|y\rangle$: **answer / target qubit**

If you want, I can show how this oracle behaves when x is in superposition—that's where the real quantum magic starts.

Special trick (very important) Set y\=0y = 0y\=0: * |0⟩|0⟩→|0⟩|f(0)⟩|0\\rangle|0\\rangle \\rightarrow |0\\rangle|f(0)\\rangle|0⟩|0⟩→|0⟩|f(0)⟩ * |1⟩|0⟩→|1⟩|f(1)⟩|1\\rangle|0\\rangle \\rightarrow |1\\rangle|f(1)\\rangle|1⟩|0⟩→|1⟩|f(1)⟩ 👉 So the second qubit directly becomes **the function output**.

didnt understand this

Got it—let's strip this down to the simplest possible picture.

🎯 What does the oracle actually do?

It performs this operation:

$$|x\rangle|y\rangle \to |x\rangle|y \oplus f(x)\rangle$$

👉 Meaning:

- First qubit x: **unchanged**
- Second qubit y: becomes $y \oplus f(x)$ (XOR)

⚡ Now the "trick": set $y = 0$

Why 0? Because XOR with 0 is special:

- $0 \oplus a = a$

🔍 Case-by-case

Case 1: $x = 0$

$$|0\rangle|0\rangle \to |0\rangle|0 \oplus f(0)\rangle$$

Since $0 \oplus f(0) = f(0)$:

👉 Output:

$$|0\rangle|f(0)\rangle$$

Case 2: $x = 1$

$$|1\rangle|0\rangle \to |1\rangle|0 \oplus f(1)\rangle$$

Again, $0 \oplus f(1) = f(1)$:

👉 Output:

$$|1\rangle|f(1)\rangle$$

🧠 What just happened?

By choosing $y = 0$, the second qubit becomes:

👉 **exactly the function output**

So the oracle is secretly doing:

> "Keep x, and write $f(x)$ into the second qubit"

💡 Intuition (super simple)

Think of it like:

- First wire = input
- Second wire = empty slot (0)
- Oracle = fills that slot with the answer

🚨 Why not just output $f(x)$ directly?

Because quantum circuits must be **reversible**.

If we did:

$$|x\rangle \to |f(x)\rangle$$

we would **lose information about x** ❌

So instead:

$$|x\rangle|y\rangle \to |x\rangle|y \oplus f(x)\rangle$$

keeps everything reversible ✅

If you want, I can show a tiny truth table example (like $f(x) = x$ or NOT) — that usually makes it click instantly.

If you want, I can show how this oracle behaves when xxx is in superposition—that's where the real quantum magic starts.

yes

Perfect—this is where it finally *clicks*. Let's use a tiny example.

🎯 Pick a simple function

Let's take:

$$f(x) = x$$

So:

- $f(0) = 0$
- $f(1) = 1$

⚙️ Oracle rule (same as before)

$$|x\rangle|y\rangle \to |x\rangle|y \oplus f(x)\rangle$$

🧪 First: normal inputs (no quantum magic yet)

Start with $y = 0$:

- $|0\rangle|0\rangle \to |0\rangle|0\rangle$
- $|1\rangle|0\rangle \to |1\rangle|1\rangle$

👉 Works like a normal function call.

🚀 Now the REAL part: superposition

Instead of giving just 0 or 1, we input:

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

So full input:

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle$$

⚡ Apply oracle to BOTH at once

Quantum rule: it acts on each part separately.

So:

$$\frac{1}{\sqrt{2}}(|0\rangle|0\rangle \ + \ |1\rangle|0\rangle)$$

After oracle:

- $|0\rangle|0\rangle \to |0\rangle|0\rangle$
- $|1\rangle|0\rangle \to |1\rangle|1\rangle$

👉 Final state:

$$\frac{1}{\sqrt{2}}(|0\rangle|0\rangle + |1\rangle|1\rangle)$$

🤯 What just happened?

In **one single operation**, the oracle computed:

- $f(0)$
- $f(1)$

👉 simultaneously

🧠 Why this is powerful

Classically:

- You need 2 calls $\to f(0), f(1)$

Quantum:

- **1 oracle call** $\to$ both are encoded in the state

💡 Intuition

Think of it like:

You asked the oracle: “what is f(0) AND f(1)?”
And it answered both at once.

🚨 Important note

You can’t just *read both answers directly* (measurement collapses it).
But clever algorithms (Deutsch, Shor) extract useful info from this combined state.

If you want, next step I’ll show the **phase trick (y = |−⟩)** — that’s the real magic behind Deutsch’s algorithm.

Exercise 7.1. Consider a classical oracle $f(x) = \bar{x}$, where x is a bit, and $\bar{x}$ is the NOT of x. We want to turn this into a quantum oracle U_f that acts according to

$$|x\rangle|y\rangle \xrightarrow{U_f} |x\rangle|y \oplus f(x)\rangle.$$

Answer the following questions about this operator:
(a) What is the truth table of the quantum oracle?

x	y	x	$y \oplus f(x)$
0	0	?	?
0	1	?	?
1	0	?	?
1	1	?	?

(b) Is the operation reversible?
(c) What is the quantum oracle as a 4×4 matrix?
(d) Verify that the matrix is unitary.

answer (a)

We’re given $f(x) = \bar{x}$ (NOT of x) and the oracle rule:

$$|x\rangle|y\rangle \to |x\rangle|y \oplus f(x)\rangle$$

🔹 Step-by-step truth table

Compute $f(x)$ first:

- $f(0) = 1$
- $f(1) = 0$

Now evaluate $y \oplus f(x)$:

x	y	f(x)	y ⊕ f(x)	Output $|x\rangle|y \oplus f(x)\rangle$
0	0	1	1	$|0\rangle|1\rangle$
0	1	1	0	$|0\rangle|0\rangle$
1	0	0	0	$|1\rangle|0\rangle$
1	1	0	1	$|1\rangle|1\rangle$

✅ Final Answer (a)

$$
\begin{aligned}
|0\rangle|0\rangle &\to |0\rangle|1\rangle \\
|0\rangle|1\rangle &\to |0\rangle|0\rangle \\
|1\rangle|0\rangle &\to |1\rangle|0\rangle \\
|1\rangle|1\rangle &\to |1\rangle|1\rangle
\end{aligned}
$$

💡 Insight:

- When $x = 0$, $f(x) = 1 \to$ **y flips**
- When $x = 1$, $f(x) = 0 \to$ **y stays same**

answer b

✅ (b) Is the operation reversible?

Yes, it is reversible.

🧠 Why?

The oracle does:

$$(x, y) \to (x,\ y \oplus f(x))$$

- x is **unchanged**
- XOR has a special property:

$$(y \oplus f(x)) \oplus f(x) = y$$

👉 So if you apply the same operation again, you get back the original input.

🔁 Reverse operation

$$(x,\ y \oplus f(x)) \to (x,\ (y \oplus f(x)) \oplus f(x)) = (x, y)$$

⚡ Intuition

- Nothing is lost (we still have x)

- XOR is **undoable**
- So the mapping is **one-to-one**

🎯 Conclusion

✔️ The oracle is **reversible**
✔️ Hence it is a valid **quantum (unitary) operation**

answer c

✅ (c) 4×4 matrix of the oracle

From part (a), the mapping is:

$$\begin{aligned} |00\rangle &\to |01\rangle \\ |01\rangle &\to |00\rangle \\ |10\rangle &\to |10\rangle \\ |11\rangle &\to |11\rangle \end{aligned}$$

🧱 Basis order

Use standard order:

$$|00\rangle,\ |01\rangle,\ |10\rangle,\ |11\rangle$$

⚙️ Build the matrix

Each column = output of that input basis vector.

$$U_f = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

💡 Intuition

- Top-left 2×2 block = NOT acting on y when $x = 0$
- Bottom-right = identity (nothing happens when $x = 1$)

That's your oracle matrix.

answer d

✅ (d) Verify the matrix is unitary

We check:

$$U_f^\dagger U_f = I$$

🔹 Key observation

$$U_f = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- All entries are real ⇒ $U_f^\dagger = U_f^T$
- Matrix is symmetric ⇒ $U_f^T = U_f$

👉 So:

$$U_f^\dagger = U_f$$

🔁 Multiply

$$U_f^\dagger U_f = U_f U_f$$

Apply twice:

- First swap (top block), second swap → back to original
- Identity part stays identity

👉 Result:

$$U_f U_f = I$$

🎯 Final conclusion

$$U_f^\dagger U_f = I$$

✔ Matrix is **unitary**
✔ Valid quantum operation

7.1.4 Phase Oracle

If we query a quantum oracle the standard way described above, the input qubit $|x\rangle$ is unchanged while the answer qubit $|y\rangle$ becomes $|y \oplus f(x)\rangle$. There is a way to query the quantum oracle, however, that causes the answer qubit $|y\rangle$ to be unchanged while multiplying $|x\rangle$ by a phase. It works by setting $|y\rangle = |-\rangle$, which can be done by initializing the answer qubit to $|0\rangle$, applying X to turn it into $|1\rangle$, and then applying H to turn it into $|-\rangle$. That is, writing both the input and answer qubits,

$$|x\rangle|0\rangle \xrightarrow{I \otimes X} |x\rangle|1\rangle \xrightarrow{I \otimes H} |x\rangle|-\rangle.$$

Note it is possible to prepare the answer qubit $|y\rangle$ in the state $|-\rangle$ because the state of a qubit can be a superposition of $|0\rangle$ and $|1\rangle$; with a classical answer bit, this would be impossible. Now, let us expand $|x\rangle|-\rangle$ and see what happens when we query the oracle:

$$\begin{aligned}
|x\rangle|-\rangle &= |x\rangle \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \\
&= \frac{1}{\sqrt{2}} (|x\rangle|0\rangle - |x\rangle|1\rangle) \\
&\xrightarrow{U_f} \frac{1}{\sqrt{2}} (|x\rangle|0 \oplus f(x)\rangle - |x\rangle|1 \oplus f(x)\rangle) \\
&= \begin{cases} \frac{1}{\sqrt{2}} (|x\rangle|0\rangle - |x\rangle|1\rangle), & f(x) = 0 \\ \frac{1}{\sqrt{2}} (|x\rangle|1\rangle - |x\rangle|0\rangle), & f(x) = 1 \end{cases} \\
&= \begin{cases} |x\rangle|-\rangle, & f(x) = 0 \\ -|x\rangle|-\rangle, & f(x) = 1 \end{cases} \\
&= (-1)^{f(x)} |x\rangle|-\rangle.
\end{aligned}$$

We can interpret this as the answer qubit staying in the $|-\rangle$ state while the input qubit goes from $|x\rangle$ to $(-1)^{f(x)}|x\rangle$. That is, the input qubit acquires a phase. This is called *phase kickback*. Often, we drop the answer qubit, since it stays in the $|-\rangle$ state, and only write the input qubit:

We can interpret this as the answer qubit staying in the $|-\rangle$ state while the input qubit goes from $|x\rangle$ to $(-1)^{f(x)}|x\rangle$. That is, the input qubit acquires a phase. This is called *phase kickback*. Often, we drop the answer qubit, since it stays in the $|-\rangle$ state, and only write the input qubit:

$$|x\rangle \xrightarrow{U_f} (-1)^{f(x)}|x\rangle.$$

This is called a *phase oracle*, where the qubit $|x\rangle$ is multiplied by a phase $(-1)^{f(x)}$. The phase oracle will be very useful for the oracular problems we will cover.

This section looks scary, but there's **one simple idea hiding inside**.

🎯 Goal of phase oracle

Instead of storing the answer in y, we want:

$$|x\rangle \to (-1)^{f(x)}|x\rangle$$

👉 No extra output
👉 Just a **sign change (phase)**

⚡ The trick (most important line)

Prepare:

$$|y\rangle = |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

🔥 What the oracle does now

Normally:

$$|x\rangle|y\rangle \to |x\rangle|y \oplus f(x)\rangle$$

But when $y = |-\rangle$, something special happens:

👉 XOR with $f(x)$ doesn't change the state…
👉 it only introduces a **minus sign**

🧪 Result (final effect)

$$|x\rangle|-\rangle \to (-1)^{f(x)}|x\rangle|-\rangle$$

🧠 Meaning

- If $f(x) = 0 \to$ no change
- If $f(x) = 1 \to$ multiply by -1

So:

$$|x\rangle \to (-1)^{f(x)}|x\rangle$$

💡 Intuition (super simple)

Instead of writing the answer:

> we "hide" it as a **sign**

- $+ \to$ means 0
- $- \to$ means 1

🚀 Why this is powerful

Quantum algorithms don't need the actual value
👉 they use **interference of signs**

That's how:

- Deutsch's algorithm works
- Shor's algorithm works
- Grover's search works

⚠️ One key takeaway

> Normal oracle → writes answer in a qubit
> Phase oracle → encodes answer as a **phase**

If you want, I can show a **1-line shortcut proof** of why
$|-\rangle$ causes this "phase kickback" (it's actually very clean).

If you want, I can show a **1-line shortcut proof** of why |−⟩|-\\rangle|−⟩ causes this “phase kickback” (it’s actually very clean).

yes also explain what is phase kickback

Let’s do the **clean 1-line proof** first—then the intuition behind *phase kickback*.

⚡ 1-line proof (the magic)

Start with:

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Apply oracle:

$$|y\rangle \to |y \oplus f(x)\rangle$$

So:

$$|-\rangle \;\to\; \frac{1}{\sqrt{2}}\big(|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle\big)$$

Now check both cases:

- If $f(x) = 0$:

$$|0\rangle - |1\rangle = |-\rangle$$

- If $f(x) = 1$:

$$|1\rangle - |0\rangle = -(|0\rangle - |1\rangle) = -|-\rangle$$

🎯 Final result

$$|-\rangle \to (-1)^{f(x)}|-\rangle$$

So:

$$|x\rangle|-\rangle \to (-1)^{f(x)}|x\rangle|-\rangle$$

🧠 What is *phase kickback*?

Here’s the key phenomenon:

The oracle **acts on the second qubit**,
but the **effect appears on the first qubit**.

🔥 Why it's called "kickback"

- The operation flips/changes y
- But because $y = |-\rangle$, that change becomes a **sign**
- That sign multiplies the **entire state**

👉 So the effect gets "kicked back" to x

💡 Intuition (very clear version)

Think:

- $y = |-\rangle$ is a **fragile balance**:
  $|0\rangle - |1\rangle$
- XOR flips:
  - swaps $|0\rangle \leftrightarrow |1\rangle$
- Swapping inside a subtraction:

$$(|0\rangle - |1\rangle) \to (|1\rangle - |0\rangle) = -(|0\rangle - |1\rangle)$$

👉 That **minus sign appears outside**
👉 affecting the whole system

🎯 Final takeaway

- Oracle normally writes: $f(x)$ into y
- With $|-\rangle$: it writes **nothing**, but adds a **phase**

$$|x\rangle \to (-1)^{f(x)}|x\rangle$$

🚀 Why this matters

This tiny effect powers:

- **Deutsch algorithm** → detect constant vs balanced
- **Grover** → mark solutions with phase
- **Shor** → interference patterns

If you want, I can show a **2-line shortcut trick** to remember this forever (no derivation needed).

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yes

Here's the **2-line shortcut**—the one people actually remember:

⚡ The Trick

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$X|-\rangle = -|-\rangle$$

🎯 Why this solves everything

- Oracle does: **apply** X **if** $f(x) = 1$
- If $f(x) = 0$: nothing happens
- If $f(x) = 1$: apply X

Now use the trick:

- $f(x) = 0 \to$ no $X \to$ same state
- $f(x) = 1 \to X|-\rangle = -|-\rangle$

🔥 Final result instantly

$$|x\rangle|-\rangle \to (-1)^{f(x)}|x\rangle|-\rangle$$

🧠 One-line memory

> "$|-\rangle$ turns an X gate into a **minus sign**."

💡 That's phase kickback

- Oracle tries to flip y (apply X)
- But $y = |-\rangle$ converts that flip into a **phase**
- That phase affects the whole system → shows up on x

If you remember just this:

> **"X on |−⟩ = −|−⟩"**

you can derive the entire phase oracle in seconds.

Exercise 7.3. Quantum oracles are quantum gates, so they act across superpositions. Consider an input qubit in the superposition state

$$\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle,$$

and an answer qubit in the state $|-\rangle$. Show that the quantum oracle acts by

$$\left(\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle\right)|-\rangle \xrightarrow{U_f} \left(\frac{\sqrt{3}}{2}(-1)^{f(0)}|0\rangle + \frac{1}{2}(-1)^{f(1)}|1\rangle\right)|-\rangle.$$

Use the one fact you just learned:

|x\rangle|-\rangle ;\xrightarrow{U_f}; (-1)^{f(x)} |x\rangle|-\rangle
]

🎯 Apply it term-by-term

Input:

$$\left(\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle\right)|-\rangle$$

Quantum rule: **linearity** → apply oracle to each part.

🔹 First term

$$\frac{\sqrt{3}}{2}|0\rangle|-\rangle \;\rightarrow\; \frac{\sqrt{3}}{2}(-1)^{f(0)}|0\rangle|-\rangle$$

🔹 Second term

$$\frac{1}{2}|1\rangle|-\rangle \;\to\; \frac{1}{2}(-1)^{f(1)}|1\rangle|-\rangle$$

⚡ Combine

$$\left(\frac{\sqrt{3}}{2}(-1)^{f(0)}|0\rangle + \frac{1}{2}(-1)^{f(1)}|1\rangle\right)|-\rangle$$

✅ Final Answer

$$U_f\left(\left(\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle\right)|-\rangle\right) = \left(\frac{\sqrt{3}}{2}(-1)^{f(0)}|0\rangle + \frac{1}{2}(-1)^{f(1)}|1\rangle\right)|-\rangle$$

💡 Insight (important)

- Each basis state $|x\rangle$ picks up its own phase
- Coefficients stay same, only **signs change**

👉 That's exactly what the question wanted you to show.

Exercise 7.4. We saw that when the answer qubit is in the state $|-\rangle$, we get phase kickback. Let us explore what happens if the answer qubit is in the state $|+\rangle$. Suppose an input qubit and answer qubit are in the state $|x\rangle|+\rangle$, where x is a bit. If we apply the quantum oracle U_f to this, which maps $|x\rangle|y\rangle$ to $|x\rangle|y \oplus f(x)\rangle$, what do we get if

(a) $f(x) = 0$?
(b) $f(x) = 1$?
(c) How do your answers in parts (a) and (b) compare to the initial state $|x\rangle|+\rangle$?

Key identity to use:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad X|+\rangle = |+\rangle$$

👉 Unlike $|-\rangle$, applying X does **nothing** to $|+\rangle$.

(a) If $f(x) = 0$

Oracle does nothing:

$$|x\rangle|+\rangle \to |x\rangle|+\rangle$$

(b) If $f(x) = 1$

Oracle applies X to second qubit:

$$|x\rangle|+\rangle \to |x\rangle X|+\rangle = |x\rangle|+\rangle$$

(c) Comparison with initial state

In **both cases**:

$$|x\rangle|+\rangle \to |x\rangle|+\rangle$$

👉 **No change at all**

🎯 Final insight

- With $|-\rangle$: phase kickback happens
- With $|+\rangle$: nothing happens

🧠 One-line memory

> $|-\rangle$ turns X into a **phase**
> $|+\rangle$ makes X **invisible**

That's why $|-\rangle$ is used in quantum algorithms—not $|+\rangle$.